Assignment 5: Database Creation and Table Implementation

SQL DDL and table relationships

Project Topic: Hotel Management System

1. Objective

Create a database for the Hotel Management System and implement the eight tables from the finalized design. The SQL below uses MySQL 8 syntax.

2. Database and Table Creation (DDL)

Run the following script in MySQL Workbench or another MySQL client. Create parent tables before tables that refer to them through foreign keys.

```
CREATE DATABASE hotel_management;
USE hotel_management;

CREATE TABLE Guests (
  guest_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(100) UNIQUE,
  phone VARCHAR(15) NOT NULL,
  address VARCHAR(150),
  id_proof_type VARCHAR(30),
  id_proof_number VARCHAR(30) UNIQUE
);

CREATE TABLE RoomTypes (
  room_type_id INT PRIMARY KEY,
  type_name VARCHAR(30) NOT NULL UNIQUE,
  base_price DECIMAL(10,2) NOT NULL CHECK (base_price >= 0),
  max_occupancy INT NOT NULL CHECK (max_occupancy > 0),
  description VARCHAR(200)
);

CREATE TABLE Rooms (
  room_id INT PRIMARY KEY,
  room_number VARCHAR(10) NOT NULL UNIQUE,
  floor_number INT NOT NULL CHECK (floor_number >= 0),
  room_type_id INT NOT NULL,
  status VARCHAR(20) NOT NULL DEFAULT 'Available'
    CHECK (status IN ('Available','Occupied','Under Maintenance')),
  FOREIGN KEY (room_type_id) REFERENCES RoomTypes(room_type_id)
);

CREATE TABLE Staff (
  staff_id INT PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  designation VARCHAR(40) NOT NULL,
  phone VARCHAR(15) NOT NULL,
  shift VARCHAR(20) NOT NULL DEFAULT 'Morning'
    CHECK (shift IN ('Morning','Evening','Night'))
);

CREATE TABLE Reservations (
  reservation_id INT PRIMARY KEY,
  guest_id INT NOT NULL,
  room_id INT NOT NULL,
```

```
  staff_id INT NOT NULL,
  check_in_date DATE NOT NULL,
  check_out_date DATE NOT NULL,
  num_guests INT NOT NULL CHECK (num_guests > 0),
  booking_status VARCHAR(20) NOT NULL DEFAULT 'Confirmed'
    CHECK (booking_status IN ('Confirmed','Checked-In','Checked-Out','Cancelled')),
  booked_room_rate DECIMAL(10,2) NOT NULL CHECK (booked_room_rate >= 0),
  CHECK (check_out_date > check_in_date),
  FOREIGN KEY (guest_id) REFERENCES Guests(guest_id),
  FOREIGN KEY (room_id) REFERENCES Rooms(room_id),
  FOREIGN KEY (staff_id) REFERENCES Staff(staff_id)
);

CREATE TABLE Payments (
  payment_id INT PRIMARY KEY,
  reservation_id INT NOT NULL,
  amount DECIMAL(10,2) NOT NULL CHECK (amount >= 0),
  payment_date DATE NOT NULL,
  payment_mode VARCHAR(20) NOT NULL
    CHECK (payment_mode IN ('Cash','Card','UPI','Net Banking')),
  payment_status VARCHAR(20) NOT NULL DEFAULT 'Pending'
    CHECK (payment_status IN ('Paid','Pending','Refunded')),
  FOREIGN KEY (reservation_id) REFERENCES Reservations(reservation_id)
);

CREATE TABLE Services (
  service_id INT PRIMARY KEY,
  service_name VARCHAR(50) NOT NULL UNIQUE,
  price DECIMAL(10,2) NOT NULL CHECK (price >= 0)
);

CREATE TABLE ServiceOrders (
  service_order_id INT PRIMARY KEY,
  reservation_id INT NOT NULL,
  service_id INT NOT NULL,
  quantity INT NOT NULL DEFAULT 1 CHECK (quantity > 0),
  order_date DATE NOT NULL,
  FOREIGN KEY (reservation_id) REFERENCES Reservations(reservation_id),
  FOREIGN KEY (service_id) REFERENCES Services(service_id)
);
```

3. Table Relationships

Parent table	Child table	Relationship
RoomTypes	Rooms	One room type can be used by many rooms
Guests	Reservations	One guest can have many reservations
Rooms	Reservations	One room can have reservations on different dates
Staff	Reservations	One staff member can process many reservations
Reservations	Payments	One reservation can have multiple payment records
Reservations	ServiceOrders	One reservation can have many service orders
Services	ServiceOrders	One service can appear in many service orders

4. Verify the Tables

```
SHOW TABLES;\nDESCRIBE Guests;\nDESCRIBE Reservations;\nSHOW CREATE TABLE ServiceOrders;
```

Expected result: SHOW TABLES should list Guests, RoomTypes, Rooms, Staff, Reservations, Payments, Services and ServiceOrders. DESCRIBE displays columns, data types, nullability and key information. SHOW CREATE TABLE displays the full table definition including constraints.

5. Conclusion

The DDL script creates the database structure and defines primary keys, foreign keys and column types. The constraints are explained and tested in Assignment 6.